

PITTSBURNT 2.7

UI and Demo Revision Report

Audit basis: latest delivered pittsburnt-main (3).zip - Scope: the ten approved UI and demo revisions only - Gemini API and Add New Solution work excluded

Conclusion

The engine passes all 56/56 property checks. The current demo weakness is not primarily the math; it is perceived pedestrian density, agent contrast, panel collisions, and metric clarity. Version 2.7 keeps the exposure calculation intact while separating physical simulation time from presentation timing, so more people are visible and their movement is slower and easier to read.

Area	Verified current state	2.7 decision
People	`AGENT_COUNT = 50`; `DEMO_TIME_COMPRESSION = 15`	Separate 70x flow cadence, slow visual movement, and a 72-agent visible target
Initial layers	`NO_LAYERS.agents = true`	Bug: start every layer at false
Agent contrast	White stroke exists inside the sprite, but disappears at small icon sizes	Increase size and add an independent backplate/drop shadow
Easter egg	MapView contains zoom>=16 inspect, sweat, and unit-click behavior	Remove completely
Setup order	Time -> Who -> Future -> Existing Protection	Move Existing Protection to the first section
Result ROI	Unspent budget is a primary PlanRow	Replace with exposure-reduction efficiency
Provenance	`SOU / COM / ASS / MIS`	Use friendly full-word labels
Severe	Person-minutes only; no people count or share	Keep definition; add visible severe agents X/N and percent
Panels	Setup 332px; Result 356px; StageOverlay cannot minimize	Setup 292px; Result 320px; center safe zone; all persistent panels minimize

Critical interpretation

- The demo looks sparsely populated at 15x because of the **50-agent cap and weak contrast**, not because of compression itself. Raising only the multiplier to 70x would merely make movement faster.
- Version 2.7 therefore does not use one compression constant for everything. Arrival cadence and on-screen travel duration become separate presentation parameters.
- Severe Exposure is not a headcount. It is accumulated walking plus waiting person-minutes at `UTCI >= 38 C`.

1. Pedestrian Flow and Visibility

1) Separate 70x flow from slow movement

Use `FLOW\_TIME\_COMPRESSION = 70` and `VISIBLE\_AGENT\_TARGET = 72`. Increase pipeline display routes from 55 to at least 90 per combination. Cap the separate visual duration so Older Adults do not appear to glide rapidly across the map.

```
// agents.ts - presentation clocks, never exposure math
export const FLOW_TIME_COMPRESSION = 70;
export const VISIBLE_AGENT_TARGET = 72;

function visualTravelDuration(routeMinutes, persona) {
  const base = (routeMinutes * 60_000 / FLOW_TIME_COMPRESSION) * 4;
  const minimum = persona === "mobility_constrained" ? 12_000
    : persona === "older_adults" ? 10_000 : 7_500;
  return clamp(base, minimum, 18_000);
}
// The engine still uses physical routeMinutes with no compression.
```

Acceptance: At the default demo zoom on a 1366x768 display, an average of 60-80 agents is visible. A representative 150 m Older Adult trip remains visible for at least about 10 seconds and reads slower than a Student trip. Zoom changes neither the route nor visual duration.

2) Increase agent contrast

- Keep the existing white sprite stroke, but enlarge icon sizes by roughly 25-35% from zoom 12 through 17.
- Add an `agent-shadow` circle layer directly below `agent-dot`: 0.92-opacity white backplate, 1.2 px slate stroke, and a soft black shadow.
- Render agents above road layers. Give a red agent one brief outer pulse; retain the cumulative severe halo separately.

```
agent-shadow: circle-radius [z12: 4, z15: 7, z17: 10]
circle-color: rgba(255,255,255,.92)
circle-stroke-color: rgba(15,23,42,.45)
circle-stroke-width: 1.2
agent icon-size: [z12: .42, z15: .68, z17: .92]
```

3) Remove the Easter egg

Remove the MapView `Inspect` state, zoom>=16 interaction, sweat treatment, agent freeze, unit-highlight click handler, and guide pill. Do not retain click behavior that competes with the core Crash -> Adjust story. Keep the street-segment popup under the shared contextual-popup rules.

2. Empty Start State and Setup Order

4) Move Existing Protection to the top

Place `Existing Protection` immediately below the brand header, followed by Time, Who, and Future Scenario. The first interaction becomes a satisfying map reveal as the user turns on Trees, Canopy, and Building Shade one at a time.

```
SETUP ORDER
1. Existing Protection
2. Time
3. Who
4. Future Scenario
5. CRASH TEST
```

5) Fully empty initial state and Moving People

The direct cause of the current mismatch is `NO\_LAYERS.agents = true`: the button appears selected, but MapView cannot draw agents because Who and Time are still null.

```
const NO_LAYERS = {
  shadow: false,
  canopy: false,
  trees: false,
  buildings: false,
  trips: false,
  agents: false, // current code is true; fix this
};
```

- On first entry and after Reset, every protection pill is unpreserved and every dynamic map layer is empty or hidden.
- Disable Moving People until Who + Time are selected. If activated early, show "Choose Who and Time first" and focus those selectors.
- After Who + Time are selected, turning on Moving People immediately shows neutral agents before the Crash Test.
- Track `mapReady` and `tripDataReady` as state/callbacks rather than refs only, so agent rendering reruns when trip GeoJSON finishes loading.

Initial/Reset acceptance matrix

State	Pills	Map	Result/Budget
First load	All OFF	Context basemap only	None
Tree click	Tree only ON	Existing trees only	None
Who + Time + People	People ON	Neutral agents immediately	None
Crash complete	Selections retained	Agents + thermal result	Shown
Reset	All OFF	Dynamic source 0; feature-state 0	None

3. Result Metrics and Provenance Labels

6) Show investment efficiency, not remaining budget

Use **Protection Efficiency = Severe person-minutes avoided per \$10,000** as the primary result. It matches the optimizer primary metric and explains policy value more directly than "the plan left \$1,000." The per-dollar value is too small to read well and belongs in a tooltip.

```
severe_saved = before_severe - after_severe
efficiency_per_dollar = severe_saved / spent_usd
efficiency_per_10k = efficiency_per_dollar * 10 000
```

Verified current snapshot: For 2035, 3 PM, Older Adults, Balanced, and a \$250K budget, the plan spends \$249,000 on 170 Trees + 3 Shelters. Severe Exposure falls from 949.83 to 843.80, saving 106.03 person-minutes. Efficiency is 0.000426 per \$1, or 4.26 severe person-minutes avoided per \$10K. This value is specific to dataset 6a0955d7 and the current cost/model version; it must never be hard-coded.

Recommended UI conv:

Protection efficiency 4.26 severe person-min avoided / \$10K
Higher is better - based on this modeled run

7) Replace the `ASS` badge

In English, `ASS` is likely to be read as slang before "assumption," so it is not suitable for the demo. Replace every abbreviated `SOU` / `COM` / `ASS` / `MIS` tag with a friendly full label.

Current	2.7 label	Color	Meaning
SOU	Source	Blue	Value taken from an external source
COM	Calculated	Slate	Value calculated by the engine
ASS	Assumed	Amber	Explicit planning assumption
MIS	Missing	Red	Required value not yet available

8) Separate Severe Exposure from people count

`Severe Exposure` is accumulated person-minutes at UTCI 38 C or above, not the number of people. Translating the current result as "950 people" would be incorrect. Show the duration metric together with a clearly separate on-screen agent metric.

```
Severe exposure time      950 person-min
Visible agents in severe heat  18 / 50 (36%)

visibleSevere = currentFrameAgents.filter(a => a.utci >= 38).length
visibleRate = visibleSevere / visibleAgentCount
```

- The tooltip must state that `Visible agents` is the share of representative icons in the current frame, not a Pittsburgh population risk rate.
- Update with a 500 ms moving average or one-second ticks so the number does not flicker on every animation frame.
- A whole modeled-trip share requires full per-trip segment membership. Do not claim a population-wide rate from the current aggregate person-minute arrays and up to 55 display routes.

4. Panel Layout and Collision Prevention

9) Narrow the panels, minimize Plan Complete, and reserve zones

Surface	Current	2.7 recommendation	Placement rule
Setup panel	332px	292px	Fixed left:16, top:16
Result panel	356px	320px	Fixed right:16, top:16
Stage / Plan Complete	max 520px; no minimize	max 460px + Minimize/Restore	Center safe zone from left:320 to right:352
Legend	left 352px	Center-safe bottom or inside Result	Never collide with CTA or panels
Modal	Full overlay	Keep	Lock background interaction while open

- Move `StageOverlay` onto the shared `Panel` behavior, or at minimum give it the same minimize state.
- When minimized, reduce it to a small top-center `Plan complete - Restore` pill.
- At 1366x768, the intersection area among Setup, Stage, and Result bounding rectangles must be zero.
- Calculate Map camera `fitBounds` padding from open-panel widths: left 308, right 336, top 16, bottom CTA 128.

10) Shared collision manager for contextual windows

Removing the Easter egg eliminates agent/tree/shelter inspect cards, but street popups and future contextual windows still need one shared rule so they never cover persistent panels.

```
uiZones = {  
  leftPanel: rect(16, 16, 292, ...),  
  rightPanel: rect(viewportW - 336, 16, 320, ...),  
  stage: centerSafeRect,  
  cta: bottomCenterRect,  
};
```

```
placePopover(anchor, preferredSides, uiZones)  
-> first candidate with zero overlap  
-> otherwise open compact bottom drawer inside centerSafeRect
```

- Allow at most one contextual popover. Clicking a new object replaces the previous popover.
- Let Mapbox flip popup anchors, but test candidate offsets/anchors so the popup opens only inside the safe area.
- Every persistent surface gets Minimize/Restore. Temporary popups close by Close or outside click.

P0 development checklist

Gate	Pass condition
Empty start	`NO_LAYERS` all false; dynamic map features 0; selected pills 0
People ready	After Who + Time, People ON immediately shows 60-80 representative agents
Motion	Separate 70x flow from slow visual duration; preserve persona ordering and zoom independence
Contrast	Agent silhouette remains identifiable on light/dark roads and red/orange thermal lines
No Easter egg	Inspect/sweat/freeze/unit-click state and guidance are absent from the bundle
Metric truth	Severe time is person-min; visible count is X/N and percent; no population-risk claim
Efficiency	Replace remaining budget row with severe person-min avoided per \$10K; matches snapshot
Panels	Zero Setup/Stage/Result overlap at 1366x768 and 1920x1080; all minimize/restore
Reset	Reset layers, pills, results, popups, feature-state, and RAF
Regression	Existing Python engine property checks remain 56/56

Excluded from this report

Gemini token/API work, Add New Solution, and custom-solution validation are being handled separately and do not block acceptance of this 2.7 patch.

Files to edit first

`web/app/page.tsx` -> empty layers and panel zones - `web/components/SetupCard.tsx` -> section order and People prerequisites - `web/lib/agents.ts` and `pipeline/step05\_trips.py` -> separate flow, movement, and count - `web/components/MapView.tsx` -> contrast and Easter egg removal - `web/components/ResultCard.tsx` / `ThreeState.tsx` -> efficiency and visible-agent metrics - `web/lib/provenance.ts` -> full badge labels - `web/components/StageOverlay.tsx` -> minimize and safe zone.